MIL-P-46124B(MR)
19 December 1978
SUPERSEDING
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5 October 1972

MILITARY SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIAL, IONOMER RESINS

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ionomer resins for injection molding, extrusion, casting, and blow molding.

1.2 Classification. The ionomer resins shall be of the following types and classes, as specified (see 6.2)

Type I - Molding resins

Class 1 - Low melt flow index
Class 2 - Medium melt flow index
Class 3 - High melt flow index

Type II - Resins for sheet, shape, and blow molding

Class 1 - Low melt flow index
Class 2 - Medium melt flow index

Type III - Resins for general purpose cast and blown thin sheet

Class 1 - Low melt flow index
Class 2 - Medium melt flow index

Type IV - Resins for extrusion coating

Class 1 - Low melt flow index
Class 2 - Medium melt flow index
Class 3 - High melt flow index

Type V - Resins for electrical applications

Class 1 - Low melt flow index
Class 2 - Medium melt flow index

Type VI - Resins for hot melts and potting

Class 1 - Very high melt flow index

FSC 9330

neficial comments (recommendations, additions, deletions) and any pertinent data which y be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums; Shipping and Storage, Steel, 55-Gallon

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 150 - Test for A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 638 - Tensile Properties of Plastics
D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1746 - Transparency of Plastic Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

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3. REQUIREMENTS

3.1 Material. The material shall consist of ionomer resins of various melt flow indexes, as specified herein, with or without specific additives such as antistat or mold release agents, antioxidants, and ultraviolet stabilizers. Reclaimed ionomer resins are permitted when they meet all requirements specified herein and the molecular weight is essentially the same as that of the unused polymer for the type and class of material specified.

3.2 Property values. The material shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3.

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Table I. Property values[1/]

Property	Type I			Type II		Type III		Type IV			Type V		Type VI
	Class 1	Class 2	Class 3	Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Class 3	Class 1	Class 2	Class 1
Melt flow index, grams per 10 minutes range	0.8-2.0	4.0-7.0	8.0-12.0	0.5-1.5	2.0-5.0	0.8-2.0	2.0-5.0	0.8-2.0	2.0-5.0	8.0-15.0	0.5-1.5	2.0-5.0	175-350
Tensile strength, MPa (psi) min.	17 (2400)	17 (2400)	17 (2400)	21 (3000)	21 (3000)	21 (3000)	15 (2200)	20 (2900)	17 (2400)	14 (2000)	17 (2400)	17 (2400)	8.3 (1200)
Elongation, percent, min.	250	200	250	240	240	250	250	240	240	200	-	-	-
Specular transmittance[2/], percent, min.	-	75	-	75	75	75	75	-	-	-	-	-	-
Dielectric constant, at 1 MHz, max.	-	-	-	-	-	-	-	-	-	-	2.40	2.40	-
Dissipation factor, at 1 MHz	-	-	-	-	-	-	-	-	-	-	0.003	0.003	-

1/ Metric abbreviations: MPa = megapascal, MHz = megahertz.

2/ Not applicable to colored material.

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3.3 Color. Unless otherwise specified by the procuring activity, the color shall be natural, which is slightly yellow to colorless.

3.4 Uniformity and form. All material shall be uniform in form and color as determined visually. The material shall be in the form of free flowing pellets.

3.5 Workmanship. The material shall be clean and free of foreign particles and other contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type and class submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of this specification and of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification will govern. A suppliers certificate of compliance with 3.1 including use of additives such as antioxidants and ultraviolet stabilizers, shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 2300 kilograms (5000 pounds) for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

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4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination specified in table II shall be approximately one pound.

Table II. Examination of the material for defects in appearance and workmanship

Examine	Defect
Appearance and workmanship	Form or color not uniform Form improper Color improper, if specified Not clean, presence of foreign material

4.2.2.1.2 Examination of the packaging. An examination shall be made in accordance with table III, to determine that packaging, packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Table III. Examination of packaging

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability
	Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers
Quantity of material	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

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4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.3 Classification of tests. All tests under this specification shall be classified as lot acceptance tests. They shall be made on each lot of material offered against this specification and, in conjunction with the above examination, shall be the basis of acceptance or rejection of the lot.

4.2.4 Testing. The material shall be tested for the characteristics listed in table I in accordance with the test methods specified herein for each lot submitted for inspection. Sampling shall be in accordance with MIL-STD-105, and the lot size for the purpose of determining sample size for testing shall be expressed in units of 450 kilograms (1000 pounds) of material. The inspection level shall be S-1. The sample size (number of units of product drawn from a lot) shall be in accordance with the single sampling plan for normal inspection and each unit shall consist of sufficient material to prepare all specimens required for testing. Each sample unit shall be tested and must meet all applicable requirements of section 3. Test results for each characteristic in each sample unit shall be the averaged results from the specimens tested for that characteristic.

4.3 Test methods.

4.3.1 Specimen preparation and testing. Test specimens shall be prepared by compression molding under conditions specified by the manufacturer. Unless otherwise specified, specimens shall be conditioned in accordance with procedure A of ASTM D 618 at 23° ± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity, and tested under these same conditions of temperature and relative humidity.

4.3.2 Melt flow index. Melt flow index shall be determined in accordance with ASTM D 1238. Three determinations shall be made using condition E and either procedure A or B of ASTM D 1238.

4.3.3 Tensile strength and elongation. Three specimens shall be tested in accordance with ASTM D 638 at speed D.

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4.3.4 Specular transmittance. Three specimens shall be tested in accordance with ASTM D 1746 using procedure A.

4.3.5 Dielectric constant and dissipation factor. Five specimens shall be tested in accordance with ASTM D 150 using a frequency of 1 megahertz. Specimen dimensions shall be 102mm (4 inches) diameter and 3.17mm (1/8 inch) thick.

5. PACKAGING

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2)

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 90 kilograms (200 pounds), maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 180 kilograms (400 pounds), maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.10 millimetre (0.004 inch) thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 90 kilograms (200 pounds) maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 180 kilograms (400 pounds), maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.10 millimetre (0.004 inch) thick polyethylene liner properly heat sealed.

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5.1.3 Commercial. Packing shall be in accordance with MIL-STD-1188.

5.2 Marking. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Ionomer resins are intended for use in products requiring impact toughness over a broad temperature range (-120° to +155°F) good resistance to stress cracking environments, good resistance to commercial solvents at ambient temperatures. good low temperature flex life, good abrasion resistance and resistance to tearing, and good optical properties.

Type I materials covers molding resins. Class 1 is intended for maximum impact toughness, class 2 for clarity and improved mold release, and class 3 for maximum flow in thin sections.

Type II material covers resins for sheet, shape and blow molding applications. Class 1 is intended for sheeting thicker than 30 mils as well as general purpose extrusion and blow molding. Class 2 is intended for heavy sheet with greater tear resistance.

Type III material covers resins for cast and blown thin sheet. Class 1 is intended for general purpose cast and blown thin theet, and class 2 for laminating to metal, wood, and cloth. Class 2 is also intended for tuoe, pipe. and shape extrusion offering greater stress crack resistance. Class 2 is offered for high tear resistant, thin sheet.

Type IV material covers resins for extrusion coating. Class 1 is intended for general purpose paper, board, and foil coatings with high resistance to oils and greases. Classes 2 and 3 also include specialty resins with such additives as slip additives, anti-blocking additives, and chill roll release agents.

Type V material covers resins for electrical applications. Class 1 is intended for electrical insulation and jacketing, and class 2 for high speed processing.

Type VI material covers resins for hot melts and potting. Hot melts are intended for fabric impregnation.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Color, if required (see 3.3).

(c) Level of packing (see 5.1).

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